

Abstract

Efficient public ferry transportation is essential for ensuring reliable mobility, passenger safety, and sustainable resource utilization within urban waterfront environments. The Toronto Island Ferry system serves as the primary transportation link between the Jack Layton Ferry Terminal and the Toronto Islands, operating under fixed vessel capacities, limited staffing resources, and highly variable seasonal demand. Although detailed ticket sales and redemption records are collected continuously, these operational data have not been fully utilized to support evidence-based decision-making regarding ferry deployment and capacity management.

This research presents a comprehensive **Ferry Capacity Utilization & Operational Efficiency Analytics System** designed to transform historical ticket activity into actionable operational intelligence. Using ferry ticket sales and redemption data collected between **2015 and 2025**, the study develops an analytical framework to evaluate utilization patterns, identify congestion-prone intervals, detect underutilized operating periods, and support strategic resource allocation. The analysis is conducted across multiple temporal resolutions, including **15-minute, hourly, daily, monthly, seasonal, and yearly** intervals, enabling a detailed understanding of operational behavior throughout the ferry network.

A structured data analytics methodology is employed, beginning with data preprocessing, timestamp standardization, quality assessment, anomaly detection, and feature engineering. Several operational metrics are derived, including **Total Activity Load, Redemption Pressure Ratio, Capacity Utilization Ratio, Operational Load Index (OLI), Congestion Pressure Index (CPI), Idle Capacity Indicator, Peak Strain Duration, and Operational Variability Score**. These indicators collectively provide quantitative measures of ferry utilization, congestion intensity, operational stability, and idle capacity across different operating conditions.

Exploratory Data Analysis (EDA) and statistical analyses are performed to investigate temporal trends, seasonal demand fluctuations, weekday versus weekend utilization differences, and operational variability. Statistical techniques, including descriptive statistics, correlation analysis, normality testing, independent sample t-tests, analysis of variance (ANOVA), confidence interval estimation, and outlier detection, are used to validate operational patterns and identify significant differences across time periods. Interactive visualizations further support the interpretation of capacity utilization, congestion behavior, and operational efficiency.

The research additionally proposes an interactive **Streamlit-based dashboard** that enables transportation managers to monitor key performance indicators in real time through dynamic filtering, timeline visualizations, congestion monitoring, seasonal comparisons, and operational scorecards. The dashboard facilitates rapid identification of peak demand periods, idle operating windows, and long-term utilization trends, thereby supporting informed scheduling and resource allocation decisions.

The findings demonstrate that operational analytics can significantly improve ferry service planning by identifying opportunities to optimize vessel deployment, reduce unnecessary operating costs during low-demand periods, improve passenger comfort through congestion management, and enhance long-term service reliability. Rather than relying solely on static

schedules or historical experience, transportation authorities can adopt data-driven strategies that respond more effectively to temporal demand variability and seasonal operational requirements.

Overall, this study illustrates how historical ticket activity data can be transformed into a comprehensive operational decision-support framework for public ferry transportation. The proposed analytics system provides practical value for the **Toronto Government Parks, Forestry & Recreation** by supporting sustainable transportation planning, operational efficiency, capacity optimization, and evidence-based policy development. The methodology is sufficiently flexible to be adapted for other public transit systems where capacity management and operational performance are critical to service delivery.

1. Introduction

1.1 Background

Public ferry transportation plays a vital role in supporting urban mobility by providing reliable, efficient, and environmentally sustainable access to island communities and recreational destinations. In Toronto, the Jack Layton Ferry Terminal serves as the primary gateway connecting the mainland to Centre Island, Hanlan's Point, and Ward's Island. Thousands of residents and visitors depend on these ferry services throughout the year for commuting, tourism, recreation, and access to public facilities. The effectiveness of ferry operations directly influences passenger satisfaction, operational costs, service reliability, and public safety.

Unlike many land-based transportation systems, ferry operations are constrained by fixed vessel capacities, scheduled departures, weather conditions, staffing limitations, and docking infrastructure. These constraints require transportation authorities to allocate resources efficiently while maintaining acceptable service levels. During peak tourist seasons and public holidays, passenger demand often exceeds average operational capacity, resulting in congestion, extended waiting times, and increased operational pressure. Conversely, during off-peak periods, ferries may operate with significantly underutilized capacity, increasing operational costs without corresponding passenger demand.

As cities increasingly adopt data-driven transportation management strategies, operational analytics has become an essential component of public transit planning. Rather than relying solely on historical schedules or managerial experience, transportation agencies now utilize historical operational data to evaluate system performance, forecast demand patterns, and optimize resource allocation. Ticket transaction data represents one of the most valuable yet underutilized sources of operational intelligence because it provides continuous information regarding passenger activity across different time intervals and seasons.

1.2 Research Context

The Toronto Island Ferry system records detailed ticket sales and redemption transactions throughout daily operations. These records capture passenger activity at regular 15-minute intervals over multiple years, creating a rich longitudinal dataset that reflects operational behavior under varying seasonal and temporal conditions. Despite the availability of these records, operational decisions have traditionally focused on predefined schedules rather than systematic analysis of utilization patterns.

Without structured analytics, several operational questions remain difficult to answer:

- Which periods consistently experience the highest operational pressure?
- When do ferries operate significantly below practical capacity?
- How does utilization vary between weekdays and weekends?
- How do seasonal tourism patterns influence ferry operations?
- Which operating windows present opportunities for cost optimization?
- How stable are operational workloads throughout the year?

Addressing these questions requires more than simple reporting. It demands an analytical framework capable of transforming raw ticket activity into measurable indicators of operational performance, efficiency, and capacity utilization.

1.3 Problem Statement

Although extensive ticket sales and redemption data are collected by the Toronto Island Ferry system, the absence of an integrated analytical framework limits their practical use for operational decision-making. Existing reporting practices primarily summarize ticket transactions without providing meaningful measures of ferry utilization, congestion intensity, idle operating periods, or long-term operational efficiency.

Consequently, transportation managers face challenges in:

- Measuring ferry capacity utilization consistently across operating periods.
- Identifying congestion-prone intervals before operational bottlenecks occur.
- Detecting persistent underutilization that increases operating costs.
- Evaluating seasonal differences in operational performance.
- Supporting evidence-based staffing and vessel deployment decisions.
- Monitoring operational efficiency through standardized performance indicators.

These limitations reduce the ability of decision-makers to optimize ferry operations while balancing service quality, passenger comfort, safety, and financial sustainability.

1.4 Research Objectives

The primary objective of this study is to develop a comprehensive operational analytics framework capable of evaluating ferry capacity utilization and operational efficiency using historical ticket activity data.

The specific objectives are:

Primary Objectives

- Quantify ferry capacity utilization using ticket sales and redemption records.
- Measure operational efficiency across multiple temporal resolutions.
- Identify congestion-prone intervals requiring additional operational attention.
- Detect underutilized operating periods representing idle capacity.

Secondary Objectives

- Compare operational performance across seasons, weekdays, and time bands.
- Develop analytical performance indicators for operational monitoring.
- Support evidence-based resource allocation and scheduling decisions.
- Improve passenger comfort by identifying recurring congestion periods.
- Provide recommendations for long-term operational optimization.

1.5 Scope of the Study

This research analyzes historical Toronto Island Ferry ticket activity spanning **2015–2025**. The dataset consists of ticket sales and ticket redemption counts recorded at **15-minute intervals**, enabling detailed operational analysis across multiple temporal dimensions.

The scope of the study includes:

- Data preprocessing and quality assessment.
- Feature engineering for operational metrics.
- Capacity utilization analysis.
- Congestion and idle-capacity assessment.
- Temporal segmentation by hour, day, month, season, and year.
- Statistical evaluation of operational behavior.
- Development of operational Key Performance Indicators (KPIs).
- Interactive dashboard development for decision support.
- Strategic recommendations for operational improvement.

The study focuses exclusively on operational efficiency derived from ticket activity. External variables such as weather conditions, vessel maintenance schedules, special events, tourism campaigns, fuel consumption, and staffing rosters are beyond the scope of the current analysis.

1.6 Significance of the Study

The proposed Ferry Capacity Utilization & Operational Efficiency Analytics System contributes to both academic research and practical transportation management.

From an operational perspective, the framework enables transportation authorities to monitor utilization continuously, identify inefficiencies, and optimize vessel deployment based on measurable evidence. Rather than relying on static schedules, decision-makers can evaluate operational performance using standardized indicators such as Capacity Utilization Ratio,

Operational Load Index, Congestion Pressure Index, Idle Capacity Percentage, Peak Strain Duration, and Operational Variability Score.

From a managerial perspective, the study supports improved budgeting, staffing, fleet scheduling, and long-term infrastructure planning. Identifying recurring congestion periods enables proactive deployment of additional ferry capacity, while recognizing consistently underutilized intervals provides opportunities to reduce unnecessary operating costs without compromising passenger service.

From an academic perspective, this research demonstrates how operational analytics techniques—including feature engineering, exploratory data analysis, statistical hypothesis testing, and interactive business intelligence—can be integrated into a unified decision-support framework for public transportation systems. The methodology is transferable to other ferry networks and public transit services where capacity management and operational efficiency are critical.

1.7 Organization of the Research

The remainder of this research paper is organized as follows:

- **Chapter 2** reviews existing literature on public transportation analytics, capacity utilization, congestion management, and operational efficiency.
- **Chapter 3** describes the dataset, research methodology, preprocessing procedures, and feature engineering techniques.
- **Chapter 4** presents the exploratory data analysis and statistical evaluation.
- **Chapter 5** discusses the operational analytics results, key findings, and performance indicators.
- **Chapter 6** provides business insights and recommendations for the Toronto Government Parks, Forestry & Recreation.
- **Chapter 7** concludes the research by summarizing findings, discussing limitations, and outlining opportunities for future work.

2. Literature Review

2.1 Introduction

Public transportation systems are fundamental to sustainable urban development, providing accessible and efficient mobility while reducing traffic congestion and environmental impacts. Among various public transport modes, ferry systems occupy a unique role in connecting waterfront communities and island destinations where road infrastructure is unavailable or impractical. Efficient ferry operations require careful coordination of vessel capacity, passenger demand, scheduling, staffing, and safety regulations. Consequently, transportation authorities increasingly rely on data analytics to improve operational performance and optimize service delivery.

This literature review examines existing research related to public transportation analytics, ferry operations, capacity utilization, operational efficiency, congestion management, and business intelligence applications. It also identifies current research gaps that motivate the development of the proposed Ferry Capacity Utilization & Operational Efficiency Analytics System.

2.2 Public Transportation Analytics

The rapid growth of digital technologies has transformed transportation management from experience-based decision-making to data-driven operational planning. Modern transit agencies routinely collect large volumes of operational data, including ticket transactions, passenger counts, vehicle locations, and service schedules. These datasets provide valuable opportunities to evaluate service performance and optimize operational strategies.

Transportation analytics combines statistical methods, data mining, machine learning, and visualization techniques to extract actionable insights from operational data. Common applications include passenger demand forecasting, route optimization, service reliability assessment, congestion detection, and infrastructure planning. Studies consistently demonstrate that analytical decision-support systems improve operational efficiency, reduce unnecessary costs, and enhance passenger satisfaction.

Despite these advancements, many transportation agencies continue to underutilize historical ticket transaction data, particularly for operational efficiency analysis. Ticket sales and redemption records are often used primarily for revenue reporting rather than for evaluating capacity utilization or identifying operational inefficiencies.

2.3 Ferry Transportation Operations

Ferry transportation differs from other public transit systems because of several operational constraints. Vessel capacity is fixed, departure schedules are less flexible than bus services, docking infrastructure limits simultaneous operations, and weather conditions may significantly influence service availability. These characteristics make efficient resource allocation particularly important.

Previous research has shown that ferry systems frequently experience significant variations in passenger demand across seasons, weekends, holidays, and special events. Tourist destinations typically exhibit sharp increases in passenger volumes during summer months, while off-season demand may decline substantially. Such fluctuations create challenges in balancing passenger comfort with operational cost efficiency.

Researchers have emphasized the importance of dynamic operational planning that adapts vessel deployment according to observed demand patterns rather than relying exclusively on predetermined schedules. Historical operational data therefore represents an essential resource for improving ferry scheduling decisions.

2.4 Capacity Utilization in Transportation Systems

Capacity utilization measures how effectively available transportation resources are used during service operations. High utilization generally indicates efficient resource deployment, while extremely high utilization may increase congestion, passenger discomfort, and safety risks. Conversely, persistently low utilization reflects idle capacity and unnecessary operational expenditure.

Transportation literature frequently evaluates capacity utilization through passenger counts, occupancy rates, seat utilization, and vehicle loading factors. These indicators help operators identify periods where service capacity does not align with passenger demand.

For ferry transportation, capacity utilization is particularly important because additional vessels cannot be deployed instantly during peak demand, and operating partially empty vessels during low-demand periods increases operational costs. Analytical measures that continuously monitor utilization provide transportation authorities with opportunities to optimize schedules while maintaining service quality.

The proposed research extends existing utilization analysis by introducing multiple complementary indicators—including the Capacity Utilization Ratio and Operational Load Index—to evaluate operational performance across different temporal scales.

2.5 Operational Efficiency Analysis

Operational efficiency refers to the ability of a transportation system to provide reliable service while minimizing unnecessary resource consumption. Efficient operations balance service availability, operational costs, passenger experience, staffing requirements, and infrastructure limitations.

Recent studies emphasize the importance of Key Performance Indicators (KPIs) for monitoring transportation efficiency. Common operational KPIs include service reliability, passenger waiting time, vehicle occupancy, schedule adherence, fleet utilization, and operational costs.

However, traditional KPIs often require multiple operational data sources that may not always be available. Ticket activity data offers an alternative perspective by capturing passenger interaction with the transportation system. When combined with feature engineering and statistical analysis, ticket records can provide meaningful indicators of operational efficiency without requiring extensive sensor infrastructure.

This study develops a structured KPI framework specifically designed for ferry operations, including Operational Load Index, Congestion Pressure Index, Idle Capacity Percentage, Peak Strain Duration, and Operational Variability Score.

2.6 Temporal Pattern Analysis

Passenger demand in transportation systems exhibits strong temporal variability influenced by time of day, day of week, season, weather conditions, holidays, and tourism activity. Temporal analysis enables transportation agencies to identify recurring operational patterns and improve scheduling decisions.

Hourly analysis typically reveals morning and afternoon demand peaks associated with commuting behavior, while weekend operations often differ significantly from weekday travel patterns. Seasonal analyses identify long-term fluctuations resulting from tourism cycles and recreational travel.

Researchers increasingly employ time-series analysis and temporal segmentation techniques to evaluate these recurring patterns. Such methods support proactive resource planning and improve the allocation of transportation assets throughout the year.

The present study applies temporal segmentation across multiple resolutions—including 15-minute intervals, hourly, daily, monthly, seasonal, and annual levels—to provide a comprehensive understanding of ferry operational behavior.

2.7 Statistical Methods in Transportation Research

Statistical analysis provides objective evidence regarding operational performance and validates observed transportation patterns. Descriptive statistics summarize operational characteristics, while inferential techniques determine whether observed differences are statistically significant.

Common statistical approaches in transportation research include:

- Descriptive statistics for summarizing operational data.
- Correlation analysis to evaluate relationships between operational variables.
- Independent sample t-tests for comparing two operational groups.
- Analysis of Variance (ANOVA) for comparing multiple operational categories.
- Confidence interval estimation for assessing statistical reliability.
- Outlier detection to identify abnormal operational events.
- Trend analysis for evaluating long-term changes in transportation demand.

These techniques enable transportation authorities to distinguish meaningful operational changes from random variation and strengthen evidence-based decision-making.

2.8 Business Intelligence and Interactive Dashboards

Business Intelligence (BI) has become an integral component of modern transportation management. Interactive dashboards provide operational managers with real-time access to key performance indicators, trend visualizations, and analytical summaries.

Visualization platforms allow decision-makers to explore operational data through interactive charts, heatmaps, scorecards, and filtering tools. Such systems improve situational awareness, accelerate decision-making, and facilitate communication between technical analysts and operational managers.

Recent advancements in open-source technologies have made it possible to develop cost-effective analytical dashboards using Python-based frameworks. Interactive applications support continuous monitoring of transportation systems and enable operational staff to investigate utilization patterns without requiring advanced programming knowledge.

In this research, a Streamlit-based dashboard is proposed to integrate analytical outputs into an intuitive decision-support environment for ferry operations.

2.9 Research Gap

Although substantial research exists on transportation analytics, several limitations remain within the context of public ferry operations.

First, many studies concentrate primarily on passenger demand forecasting rather than operational efficiency evaluation. Second, existing research often relies on multiple external data sources such as weather information, GPS tracking, or vessel sensor data, limiting applicability where such information is unavailable.

Third, relatively few studies utilize ticket sales and redemption records as the primary source for operational performance analysis. Consequently, important opportunities for evaluating capacity utilization, congestion behavior, and idle operating periods remain underexplored.

Furthermore, existing ferry studies frequently focus on isolated performance indicators rather than integrating multiple operational KPIs into a unified analytical framework capable of supporting strategic decision-making.

The proposed Ferry Capacity Utilization & Operational Efficiency Analytics System addresses these limitations by developing a comprehensive operational analytics framework based solely on historical ticket activity data. The methodology combines feature engineering, exploratory data analysis, statistical testing, KPI development, and interactive visualization to provide transportation authorities with a practical and scalable decision-support system.

2.10 Chapter Summary

The reviewed literature demonstrates that data analytics has become essential for improving transportation planning, operational efficiency, and resource allocation. Previous studies highlight the importance of capacity utilization measurement, temporal demand analysis, statistical validation, and business intelligence for supporting evidence-based transportation management.

However, a clear research gap exists regarding the systematic use of ticket sales and redemption data for ferry operational analytics. This study addresses that gap by proposing an integrated analytical framework capable of measuring utilization patterns, detecting congestion and idle capacity, evaluating operational efficiency, and supporting strategic planning for the Toronto Island Ferry system.

The following chapter presents the research methodology, dataset description, preprocessing procedures, feature engineering process, and analytical techniques used to implement the proposed Ferry Capacity Utilization & Operational Efficiency Analytics System.

3. Research Methodology

3.1 Introduction

This chapter presents the research methodology adopted to develop the **Ferry Capacity Utilization & Operational Efficiency Analytics System**. The methodology provides a structured framework for transforming historical ferry ticket transaction data into meaningful operational insights that support evidence-based decision-making. It includes the research design, dataset description, data preprocessing, feature engineering, exploratory data analysis, statistical analysis, Key Performance Indicator (KPI) development, and dashboard implementation.

The analytical workflow was designed to identify operational utilization patterns, congestion-prone intervals, idle capacity, and long-term efficiency trends within the Toronto Island Ferry system. By integrating data engineering, statistical methods, and business intelligence techniques, the proposed methodology enables transportation managers to evaluate ferry operations systematically.

3.2 Research Design

This study adopts a **quantitative, descriptive, and analytical research design**. Historical operational data are analyzed using statistical and data analytics techniques to measure ferry utilization and operational efficiency.

The research follows a sequential workflow consisting of six major phases:

1. Data acquisition and ingestion
2. Data preprocessing and quality assessment

3. Feature engineering
4. Exploratory Data Analysis (EDA)
5. Statistical analysis and KPI development
6. Interactive dashboard visualization and operational recommendations

This structured approach ensures that raw ticket activity data are progressively transformed into actionable operational intelligence.

3.3 Dataset Description

The study utilizes historical ticket transaction data from the **Toronto Island Ferry System**, covering ferry operations between **2015 and 2025**.

Each observation represents passenger activity during a **15-minute operational interval**.

Dataset Variables

Variable	Description
<code>_id</code>	Unique observation identifier
<code>Timestamp</code>	End time of each 15-minute interval
<code>Sales Count</code>	Number of ferry tickets sold during the interval
<code>Redemption Count</code>	Number of tickets redeemed during boarding

The dataset represents passenger interaction with ferry services and serves as the primary source for evaluating operational efficiency.

3.4 Data Collection

The analysis relies on historical operational records collected through the Toronto Island Ferry ticketing system. Ticket sales indicate purchasing activity, while redemption records represent passenger boarding activity.

The dataset contains regularly recorded observations at fixed 15-minute intervals, making it suitable for time-series analysis and operational performance evaluation.

No personally identifiable passenger information is included in the dataset, ensuring compliance with data privacy and ethical research standards.

3.5 Data Preprocessing

Raw operational datasets frequently contain inconsistencies that may affect analytical accuracy. Therefore, several preprocessing procedures were performed before analysis.

3.5.1 Timestamp Standardization

The **Timestamp** variable was converted into Python datetime format to enable chronological ordering and time-based aggregation.

This transformation allowed extraction of:

- Year
 - Month
 - Day
 - Hour
 - Minute
 - Weekday
 - Week Number
 - Quarter
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3.5.2 Data Cleaning

The following quality checks were conducted:

- Missing value identification
- Duplicate record detection
- Negative ticket count validation
- Chronological sorting
- Data type verification

Any invalid ticket counts were excluded from subsequent analyses to preserve data integrity.

3.5.3 Temporal Aggregation

To evaluate operational behavior across multiple resolutions, the dataset was aggregated into:

- **15-minute intervals** (original resolution)
- **Hourly summaries**
- **Daily summaries**

These aggregation levels support both detailed operational monitoring and long-term trend analysis.

3.6 Feature Engineering

Several derived variables were created to quantify operational efficiency and ferry utilization.

Total Activity Load

Represents total passenger activity during an interval.

Formula

$$\begin{aligned} &[\\ &\text{\text{Total Activity Load}} = \\ &\text{\text{Sales_Count}} + \\ &\text{\text{Redemption_Count}} \\ &] \end{aligned}$$

Redemption Pressure Ratio

Measures boarding pressure relative to ticket sales.

Formula

$$\begin{aligned} &[\\ &\text{\text{RPR}} = \\ &\frac{\text{\text{Redemption_Count}}}{\text{\text{Sales_Count}} + 1} \\ &] \end{aligned}$$

Adding one prevents division by zero.

Capacity Utilization Ratio (CUR)

CUR estimates ferry utilization efficiency by comparing redeemed tickets with tickets sold.

Higher values indicate greater utilization of available passenger demand.

Operational Load Index (OLI)

Operational Load Index is a normalized measure of interval-level operational activity.

The index ranges from 0 to 1.

Higher values indicate heavier operational demand.

Congestion Pressure Index (CPI)

The Congestion Pressure Index combines utilization and operational load into a single congestion metric.

Higher CPI values indicate periods requiring operational attention.

Idle Capacity Indicator

Intervals falling below the lower activity threshold were classified as idle-capacity periods.

These observations represent potential opportunities for cost optimization.

Peak Interval Indicator

Intervals exceeding the 90th percentile of operational activity were classified as peak operating periods.

These intervals were used to evaluate congestion duration and operational strain.

3.7 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand data characteristics before statistical modelling.

EDA included:

- Dataset overview
- Summary statistics
- Missing-value assessment
- Duplicate analysis
- Distribution analysis
- Time-series visualization
- Hourly utilization trends
- Daily activity trends
- Monthly activity
- Seasonal utilization
- Weekday vs. weekend comparisons
- Correlation analysis
- Outlier detection
- Rolling-average analysis

These analyses provided initial insights into operational patterns and informed subsequent statistical testing.

3.8 Statistical Analysis

Inferential statistical techniques were employed to evaluate operational differences and validate observed patterns.

The analyses included:

Descriptive Statistics

- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Quartiles

Correlation Analysis

Both Pearson and Spearman correlation coefficients were calculated to measure relationships among operational variables.

Normality Assessment

The Shapiro–Wilk test was performed to evaluate whether activity distributions followed a normal distribution.

Independent Sample t-Test

Operational activity was compared between weekdays and weekends.

One-Way ANOVA

Analysis of Variance (ANOVA) evaluated seasonal differences in ferry utilization.

Confidence Interval Estimation

A 95% confidence interval was calculated for average operational activity.

Outlier Detection

Two complementary techniques were used:

- Z-score analysis

- Interquartile Range (IQR)

These methods identified unusually high or low operational activity intervals.

3.9 Key Performance Indicators (KPIs)

Several operational performance metrics were developed to quantify ferry efficiency.

These include:

- Capacity Utilization Ratio (CUR)
- Operational Load Index (OLI)
- Congestion Pressure Index (CPI)
- Idle Capacity Percentage
- Peak Interval Percentage
- Peak Strain Duration
- Operational Variability Score
- Redemption Efficiency

These KPIs provide standardized measures for monitoring ferry operations over time.

3.10 Dashboard Development

An interactive dashboard was developed using the **Streamlit** framework.

The dashboard enables users to:

- Filter observations by year, season, and time band.
- Monitor operational KPIs.
- Visualize activity timelines.
- Compare seasonal performance.
- Identify congestion-prone periods.
- Detect idle-capacity intervals.
- Download filtered datasets for further analysis.

Interactive visualizations were implemented using **Plotly**, enabling dynamic exploration of operational trends.

3.11 Software and Analytical Tools

The analytical system was implemented using the Python programming language.

The primary software libraries include:

Tool	Purpose
Python	Data analysis and application development
Pandas	Data manipulation and preprocessing
NumPy	Numerical computation
Matplotlib	Statistical visualization
Plotly	Interactive visualizations
SciPy	Statistical hypothesis testing
Streamlit	Dashboard development

These open-source technologies provide a scalable and reproducible analytical environment.

3.12 Research Workflow

The complete analytical workflow adopted in this study is summarized below:

1. Historical ticket data acquisition
2. Data cleaning and preprocessing
3. Timestamp standardization
4. Temporal feature extraction
5. Feature engineering
6. Exploratory Data Analysis
7. Statistical hypothesis testing
8. KPI computation
9. Dashboard development
10. Operational interpretation and recommendations

This workflow ensures a systematic progression from raw operational records to actionable decision-support outputs.

3.13 Chapter Summary

This chapter described the methodology used to develop the Ferry Capacity Utilization & Operational Efficiency Analytics System. A structured analytical framework was established encompassing data preprocessing, feature engineering, exploratory analysis, statistical testing, KPI development, and interactive dashboard implementation.

The methodology provides a rigorous foundation for evaluating ferry operational performance using historical ticket activity data. The following chapter presents the Exploratory Data Analysis (EDA), statistical findings, and operational patterns identified through the proposed analytical framework.

4. Exploratory Data Analysis (EDA) and Statistical Analysis

4.1 Introduction

Following data preprocessing and feature engineering, an Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the Toronto Island Ferry ticket activity dataset and identify operational patterns. EDA provides an initial understanding of the distribution of ticket sales, passenger redemption activity, temporal variations, and operational load before applying statistical inference.

The objective of this chapter is to evaluate ferry operations across multiple temporal dimensions, identify congestion-prone and underutilized periods, and validate operational differences using statistical methods. Together, these analyses establish the analytical foundation for the operational insights and recommendations presented in subsequent chapters.

4.2 Dataset Overview

The processed dataset contains historical ferry ticket activity recorded at **15-minute intervals** between **2015 and 2025**. Each observation represents operational activity during a specific time interval and includes ticket sales, ticket redemptions, and several engineered operational indicators.

The preprocessing stage ensured:

- Removal of invalid observations.
- Chronological ordering of records.
- Standardized timestamp formatting.
- Extraction of temporal attributes (year, month, weekday, hour, season, and time band).
- Creation of operational efficiency metrics.

These steps ensured that the dataset was suitable for both descriptive and inferential analyses.

4.3 Descriptive Statistics

Descriptive statistics summarize the central tendency and variability of operational activity.

The following variables were examined:

- Sales Count

- Redemption Count
- Total Activity Load
- Capacity Utilization Ratio
- Operational Load Index
- Congestion Pressure Index

Measures including mean, median, minimum, maximum, standard deviation, and quartiles were calculated.

The descriptive analysis provides a quantitative overview of passenger activity and establishes baseline operational behavior across the observation period.

Key observations include:

- Ticket activity varies substantially across different time intervals.
 - Passenger boarding activity generally follows ticket purchasing trends.
 - Operational load exhibits periods of both high congestion and low utilization.
 - The distribution of activity is positively skewed, indicating that high-demand periods occur less frequently but contribute significantly to total operational workload.
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4.4 Temporal Activity Analysis

4.4.1 Fifteen-Minute Activity

The original dataset captures ferry activity at 15-minute intervals, enabling detailed monitoring of operational fluctuations.

Time-series visualization reveals:

- Regular operational cycles throughout the day.
- Concentrated periods of elevated passenger activity.
- Recurring low-demand intervals during off-peak hours.
- Seasonal fluctuations throughout the observation period.

This level of temporal resolution allows transportation managers to identify precise intervals requiring operational intervention.

4.4.2 Hourly Analysis

Hourly aggregation provides a broader understanding of operational demand.

The analysis indicates that passenger activity is not uniformly distributed throughout the day.

Operational activity generally follows three distinct phases:

- Morning build-up
- Afternoon peak operations
- Evening decline

These recurring hourly patterns support optimized staffing schedules and vessel deployment.

4.4.3 Daily Analysis

Daily aggregation smooths short-term fluctuations and highlights long-term operational trends.

Daily analysis demonstrates:

- Consistent operational cycles.
- Short-term demand fluctuations.
- Periodic spikes corresponding to high passenger activity.
- Extended periods of relatively stable operational demand.

Daily summaries provide valuable information for workforce planning and operational scheduling.

4.4.4 Monthly and Seasonal Analysis

Monthly analysis reveals substantial variation in operational activity throughout the year.

Seasonal comparisons indicate:

- Increased utilization during the summer months.
- Reduced operational demand during winter.
- Moderate activity during spring and autumn.

These findings are consistent with the seasonal recreational nature of Toronto Island transportation.

The seasonal analysis demonstrates the importance of flexible operational planning rather than maintaining identical schedules throughout the year.

4.5 Weekday and Weekend Comparison

Passenger behavior differs considerably between weekdays and weekends.

Weekend operations generally exhibit:

- Increased recreational travel.
- Higher passenger concentrations.
- Greater operational variability.

Weekday operations demonstrate:

- More stable activity patterns.
- Lower variability.
- More predictable operational demand.

These differences justify separate operational planning strategies for weekdays and weekends.

4.6 Time Band Analysis

To simplify operational interpretation, each observation was categorized into four operational periods:

- Morning
- Afternoon
- Evening
- Night

The comparison indicates that operational demand is concentrated primarily during daytime service periods.

Night operations exhibit substantially lower utilization, indicating opportunities for reducing unnecessary operational expenditure.

Time-band segmentation assists transportation managers in matching staffing levels and vessel deployment to expected passenger activity.

4.7 Distribution Analysis

Frequency distributions were examined for:

- Sales Count
- Redemption Count
- Total Activity Load

Histogram analysis indicates that operational activity does not follow a perfectly symmetric distribution.

Most observations correspond to relatively low or moderate passenger activity, while a smaller number of intervals experience exceptionally high operational demand.

Such positively skewed distributions are common within transportation systems characterized by concentrated peak periods.

4.8 Correlation Analysis

Correlation analysis was performed to evaluate relationships among operational variables.

Pearson and Spearman correlation coefficients were calculated for:

- Sales Count
- Redemption Count
- Total Activity Load
- Operational Load Index
- Capacity Utilization Ratio
- Congestion Pressure Index

The analysis demonstrates strong positive relationships between ticket sales and passenger redemption activity, indicating that increases in ticket purchases generally correspond with increased boarding demand.

Similarly, Operational Load Index and Congestion Pressure Index exhibit positive associations, confirming that higher operational activity is accompanied by increased congestion pressure.

These relationships validate the proposed feature engineering methodology.

4.9 Outlier Analysis

Operational outliers were identified using two complementary techniques:

- Z-score analysis
- Interquartile Range (IQR)

Outliers primarily represent intervals experiencing unusually high passenger activity.

Rather than treating these observations as errors, they are considered operationally significant because they correspond to periods requiring additional transportation resources.

Identifying these intervals enables transportation managers to investigate recurring congestion and evaluate scheduling improvements.

4.10 Rolling Trend Analysis

Rolling averages were calculated to smooth short-term fluctuations and reveal underlying operational trends.

The rolling analysis demonstrates:

- Stable long-term operational behavior.
- Temporary congestion spikes.
- Gradual seasonal transitions.
- Reduced influence of random short-term variation.

Rolling statistics improve interpretation by emphasizing sustained operational changes rather than isolated events.

4.11 Statistical Analysis

Inferential statistical techniques were employed to determine whether observed operational differences were statistically significant.

Normality Assessment

The Shapiro–Wilk test evaluated whether Total Activity Load followed a normal distribution.

The results indicate that operational activity deviates from normality, which is expected for transportation datasets characterized by seasonal peaks and concentrated passenger demand.

Weekday vs Weekend Comparison

An independent sample t-test compared operational activity between weekdays and weekends.

The analysis assessed whether differences in average operational load occurred due to random variation or represented statistically meaningful changes.

Where significant differences were identified, separate operational strategies for weekdays and weekends are justified.

Seasonal Comparison

One-way Analysis of Variance (ANOVA) was performed to evaluate differences among winter, spring, summer, and fall operations.

The statistical analysis demonstrates whether seasonal operational changes are sufficiently large to justify differentiated scheduling and resource allocation.

Seasonal significance supports adaptive operational planning throughout the year.

Confidence Interval Estimation

A 95% confidence interval was calculated for average operational activity.

Confidence intervals quantify the uncertainty surrounding estimated operational averages and provide decision-makers with statistically reliable performance ranges.

4.12 Operational Performance Indicators

Several analytical indicators were developed to summarize operational efficiency.

Capacity Utilization Ratio

Measures passenger utilization relative to ticket sales.

Higher values indicate greater operational utilization.

Operational Load Index

Represents normalized operational activity.

The index enables comparisons across different time periods despite varying activity levels.

Congestion Pressure Index

Combines utilization and operational load into a single operational congestion metric.

Higher values indicate periods requiring operational attention.

Idle Capacity Indicator

Identifies intervals characterized by persistently low operational activity.

These intervals represent opportunities for operational cost optimization.

Peak Interval Indicator

Classifies observations experiencing exceptionally high operational demand.

Peak intervals assist transportation managers in identifying recurring congestion windows.

4.13 Summary of Findings

The exploratory and statistical analyses reveal several important operational characteristics of the Toronto Island Ferry system.

Major findings include:

- Ferry activity exhibits clear temporal and seasonal patterns.
- Passenger demand varies considerably across hours, weekdays, and seasons.
- Congestion is concentrated within identifiable operational windows.
- Extended periods of idle capacity exist during low-demand intervals.
- Ticket sales and redemption activity are strongly correlated.
- Statistical testing confirms significant operational differences across temporal categories.
- Feature-engineered KPIs effectively summarize ferry utilization and operational efficiency.

These findings demonstrate that historical ticket activity provides sufficient information to evaluate ferry operations comprehensively and supports evidence-based resource allocation.

4.14 Chapter Summary

This chapter presented the exploratory data analysis and statistical evaluation of Toronto Island Ferry operations. Descriptive statistics, temporal analyses, distribution assessments, correlation analysis, outlier detection, rolling trend evaluation, and inferential statistical tests collectively provided a comprehensive understanding of operational behavior.

The results confirm that ferry operations exhibit measurable patterns of utilization, congestion, and idle capacity across different temporal dimensions. These analytical findings establish the basis for the operational insights, dashboard interpretation, and strategic recommendations presented in the following chapter.

5. Results and Discussion

5.1 Introduction

This chapter presents the findings obtained from the Ferry Capacity Utilization & Operational Efficiency Analytics System. The results are derived from the exploratory data analysis, feature engineering, statistical evaluation, and operational Key Performance Indicators (KPIs) developed using historical Toronto Island Ferry ticket activity data. The objective is to interpret these analytical outcomes and explain their implications for operational efficiency, resource allocation, and long-term ferry service planning.

Rather than simply reporting statistical values, this chapter examines how operational patterns influence ferry performance and identifies opportunities for improving scheduling, capacity utilization, and passenger service.

5.2 Capacity Utilization Analysis

The Capacity Utilization Ratio (CUR) provides an estimate of how effectively ferry services accommodate passenger demand based on ticket redemption activity. Analysis across the observation period indicates that utilization varies considerably over time rather than remaining constant.

Several recurring characteristics were observed:

- Capacity utilization increases significantly during high-demand periods.
- Moderate utilization is maintained during normal operating hours.
- Utilization declines substantially during off-peak periods.
- Seasonal tourism strongly influences overall utilization levels.

These findings suggest that maintaining identical ferry schedules throughout the year may not represent the most efficient operational strategy. Instead, adaptive scheduling based on observed utilization patterns could improve both operational efficiency and passenger experience.

5.3 Operational Load Assessment

The Operational Load Index (OLI) measures normalized passenger activity across all observation intervals.

The analysis reveals that operational demand follows predictable temporal cycles rather than random fluctuations. Daily operations generally exhibit gradual increases in activity during daytime service hours, followed by declining demand during evening operations.

Periods with consistently high OLI values correspond to intervals where ferry resources experience sustained operational pressure.

Conversely, prolonged periods of low operational load indicate opportunities to reduce unnecessary operating costs through optimized vessel deployment or revised service frequency.

The Operational Load Index therefore provides transportation managers with a standardized measure for comparing operational performance across different seasons and years.

5.4 Congestion Pressure Analysis

One of the primary objectives of this research was to identify intervals experiencing elevated operational pressure.

The Congestion Pressure Index (CPI) integrates operational load with passenger utilization, providing a comprehensive measure of congestion intensity.

The analysis demonstrates that congestion is concentrated within specific recurring operating windows rather than being uniformly distributed throughout the day.

Characteristics of congestion periods include:

- High passenger boarding activity.
- Elevated redemption pressure.
- Increased operational workload.
- Greater likelihood of passenger queues.
- Reduced operational flexibility.

These congestion windows represent critical periods during which additional ferry capacity or operational staff may be required.

The CPI therefore functions as an operational early-warning indicator that supports proactive resource allocation.

5.5 Idle Capacity Assessment

In contrast to congestion analysis, idle capacity analysis identifies intervals during which ferry services operate substantially below observed demand potential.

The analytical framework detected multiple recurring low-activity periods characterized by:

- Minimal ticket sales.
- Low passenger boarding activity.
- Reduced operational load.
- Sustained idle intervals.

These findings indicate that certain operating windows consistently experience underutilization.

Maintaining identical operational intensity during these periods may increase fuel consumption, staffing costs, and maintenance expenditure without corresponding passenger demand.

Consequently, idle capacity analysis provides transportation managers with evidence supporting service optimization during low-demand periods while preserving passenger accessibility.

5.6 Temporal Operational Patterns

Hourly Patterns

Hourly analysis demonstrates pronounced differences in operational activity across the day.

Passenger demand generally follows a structured operational cycle:

- Lower activity during early operating hours.
- Progressive increase during daytime.
- Peak operational pressure during busy service periods.
- Gradual reduction during late evening operations.

These recurring patterns enable transportation authorities to schedule vessels more efficiently according to anticipated passenger demand.

Weekly Patterns

Comparison between weekdays and weekends indicates meaningful operational differences.

Weekend operations generally exhibit:

- Higher recreational travel.
- Increased passenger variability.
- Greater congestion risk.

Weekday operations are comparatively more stable and predictable.

This distinction supports differentiated staffing strategies and ferry schedules based on the day of operation.

Seasonal Patterns

Seasonal analysis reveals the strongest variation within the dataset.

Summer operations consistently demonstrate elevated activity associated with tourism and recreational travel.

Winter operations exhibit considerably lower utilization.

Spring and autumn represent transitional periods characterized by moderate operational demand.

These observations confirm that ferry operations should adopt seasonal service strategies rather than uniform annual scheduling.

5.7 Statistical Interpretation

Inferential statistical analysis validates the operational observations identified during exploratory analysis.

Correlation Analysis

Strong positive relationships between ticket sales and passenger redemption activity confirm that increased purchasing behavior is closely associated with increased boarding demand.

This relationship supports the use of ticket transaction data as a reliable indicator of operational activity.

Weekday–Weekend Comparison

Independent sample t-tests demonstrate that operational activity differs significantly between weekdays and weekends.

This finding justifies separate operational planning approaches for different types of travel demand.

Seasonal Comparison

Analysis of Variance (ANOVA) indicates statistically significant differences among seasonal operational patterns.

The results reinforce the importance of adaptive scheduling and seasonal fleet management.

Outlier Interpretation

Operational outliers correspond primarily to intervals experiencing exceptionally high passenger demand.

Rather than treating these observations as data errors, they should be interpreted as valuable operational signals indicating:

- Major tourism activity.
- Holiday travel.
- Event-related demand.
- Exceptional congestion periods.

Monitoring these intervals enables transportation managers to improve preparedness for future high-demand situations.

5.8 Operational Key Performance Indicators

The analytical framework generated several KPIs that summarize ferry operational performance.

Capacity Utilization Ratio

Provides an overall measure of service utilization efficiency.

Operational Load Index

Measures interval-level operational demand using normalized activity values.

Congestion Pressure Index

Quantifies congestion intensity across operational periods.

Idle Capacity Percentage

Measures the proportion of intervals experiencing low operational activity.

Peak Strain Duration

Evaluates the persistence of high-congestion periods rather than isolated peaks.

Operational Variability Score

Measures the consistency of operational activity over time.

Collectively, these indicators provide transportation managers with a comprehensive operational monitoring framework.

5.9 Dashboard Evaluation

The Streamlit dashboard developed as part of this research integrates analytical findings into an interactive decision-support system.

The dashboard enables users to:

- Monitor operational KPIs.
- Filter analyses by year, season, and time band.
- Visualize capacity utilization trends.
- Identify congestion-prone intervals.
- Detect idle operating periods.
- Compare seasonal operational performance.
- Download filtered analytical results.

This interactive environment substantially improves accessibility compared with static reports, allowing managers to investigate operational conditions dynamically.

5.10 Discussion

The results demonstrate that historical ticket activity data can provide considerably more operational intelligence than traditional transaction reporting alone.

Instead of focusing solely on passenger counts, the proposed analytical framework evaluates operational efficiency through multiple complementary perspectives including utilization, congestion, variability, and idle capacity.

The findings indicate that ferry operations exhibit predictable temporal behavior that can support proactive operational planning.

Several important implications emerge:

- Static schedules may not fully reflect observed passenger demand.
- Seasonal scheduling offers opportunities for improved operational efficiency.
- Congestion can be anticipated through historical utilization patterns.
- Idle operating periods represent potential cost-saving opportunities.
- Interactive dashboards enhance operational transparency and support faster decision-making.

The combination of statistical analysis, feature engineering, and business intelligence therefore provides a practical foundation for evidence-based ferry management.

5.11 Chapter Summary

This chapter presented and interpreted the results of the Ferry Capacity Utilization & Operational Efficiency Analytics System. Capacity utilization, operational load, congestion pressure, idle capacity, temporal patterns, and statistical analyses collectively demonstrate that ferry operations follow measurable and recurring operational behaviors.

The developed KPIs and interactive dashboard transform historical ticket transaction data into actionable operational intelligence, enabling transportation authorities to optimize scheduling, allocate resources more effectively, and improve passenger service quality. These findings form the basis for the strategic recommendations presented in the next chapter.

6. Business Insights and Recommendations

6.1 Introduction

The primary objective of the Ferry Capacity Utilization & Operational Efficiency Analytics System is not only to analyze historical ferry ticket activity but also to convert analytical findings into practical recommendations for improving operational performance. The analyses presented in the previous chapters demonstrate that ferry operations exhibit clear temporal, seasonal, and operational patterns that can support more effective planning and resource allocation.

This chapter translates those analytical findings into actionable recommendations for **Toronto Government Parks, Forestry & Recreation**, with a focus on improving operational efficiency, reducing unnecessary costs, enhancing passenger experience, and supporting long-term transportation planning.

6.2 Key Business Insights

Insight 1: Ferry Demand Follows Predictable Temporal Patterns

The analysis indicates that ferry activity is not random but follows recurring hourly, daily, and seasonal cycles. Passenger activity consistently increases during specific operating periods and declines during others.

This predictability provides an opportunity to move from static scheduling toward dynamic, evidence-based operational planning.

Operational Implication

- Schedule vessels according to historical demand patterns rather than maintaining uniform service frequency throughout the day.

Insight 2: Seasonal Variation Strongly Influences Ferry Operations

The seasonal analysis shows substantial differences in operational activity across the year. Summer periods experience significantly higher utilization due to tourism and recreational travel, while winter periods demonstrate comparatively lower demand.

Operational Implication

- Implement season-specific operating schedules.
 - Allocate additional ferry capacity during high-demand seasons.
 - Reduce unnecessary operational intensity during low-demand months while maintaining essential service levels.
-

Insight 3: Congestion Occurs During Recurring Time Windows

The Congestion Pressure Index identifies recurring intervals with elevated operational pressure. These intervals represent periods where passenger boarding activity approaches operational limits.

Rather than responding reactively, ferry operators can anticipate these periods using historical operational intelligence.

Operational Implication

- Increase vessel frequency during recurring congestion windows.
 - Deploy additional boarding staff during peak intervals.
 - Monitor congestion KPIs continuously through the dashboard.
-

Insight 4: Persistent Idle Capacity Exists During Off-Peak Periods

Idle Capacity analysis reveals recurring periods where ferry services operate significantly below observed passenger demand.

Although maintaining continuous service remains important, excessive operational intensity during these intervals may increase operating costs without corresponding benefits.

Operational Implication

- Evaluate opportunities to reduce departure frequency during consistently underutilized periods.
 - Optimize staffing assignments during low-demand operations.
 - Reallocate operational resources toward higher-demand intervals.
-

Insight 5: Ticket Sales Effectively Reflect Operational Demand

The statistical analysis demonstrates a strong positive relationship between ticket sales and passenger redemption activity.

This finding indicates that historical ticket transaction data can serve as a reliable operational planning resource.

Operational Implication

- Integrate ticket activity into routine operational planning.
 - Develop utilization forecasts based on historical ticket behavior.
 - Use ticket trends as an early indicator of changing passenger demand.
-

6.3 Operational Recommendations

Recommendation 1: Dynamic Ferry Scheduling

Rather than applying identical schedules throughout the year, ferry operations should adopt demand-responsive scheduling.

Recommended actions include:

- Increasing departures during historically congested periods.
- Reducing unnecessary departures during prolonged low-demand intervals.
- Reviewing schedules periodically using updated operational analytics.

Expected benefits include improved utilization, reduced congestion, and lower operating costs.

Recommendation 2: Seasonal Resource Allocation

Historical analysis demonstrates clear seasonal demand variation.

Transportation authorities should therefore implement separate operational strategies for:

- Summer
- Spring
- Autumn
- Winter

Season-specific planning should include:

- Vessel allocation
- Staffing levels
- Maintenance scheduling
- Operational budgets

This approach improves both efficiency and service reliability.

Recommendation 3: Continuous KPI Monitoring

Operational efficiency should be monitored using standardized Key Performance Indicators.

Recommended KPIs include:

- Capacity Utilization Ratio
- Operational Load Index
- Congestion Pressure Index
- Idle Capacity Percentage
- Peak Strain Duration
- Operational Variability Score

Regular KPI reviews enable early identification of operational issues before they affect passenger service.

Recommendation 4: Dashboard-Based Decision Support

The Streamlit dashboard developed during this project should become part of routine operational management.

Operational managers can use the dashboard to:

- Monitor utilization trends.
- Detect congestion.
- Identify idle operating periods.
- Compare seasonal performance.
- Evaluate operational improvements over time.

Interactive analytics support faster and more informed decision-making than static operational reports.

Recommendation 5: Staffing Optimization

Passenger demand varies considerably throughout the operating day.

Historical utilization analysis should therefore guide staffing decisions.

Examples include:

- Increasing boarding personnel during peak congestion.

- Reducing staffing during extended low-demand intervals.
- Scheduling employee shifts based on operational activity rather than fixed timetables.

This strategy supports both operational efficiency and employee productivity.

Recommendation 6: Preventive Congestion Management

Rather than responding after congestion develops, transportation managers should proactively monitor congestion indicators.

Suggested actions include:

- Establishing operational thresholds for the Congestion Pressure Index.
- Triggering operational alerts when thresholds are exceeded.
- Preparing reserve vessels during historically congested periods.
- Communicating expected peak periods to passengers.

Preventive congestion management improves passenger comfort and operational resilience.

6.4 Strategic Recommendations

Long-Term Operational Planning

Historical operational analytics should be incorporated into annual transportation planning.

Long-term planning should consider:

- Tourism growth.
- Population changes.
- Infrastructure expansion.
- Climate-related operational impacts.
- Future passenger demand trends.

Historical ticket activity provides valuable evidence for these strategic decisions.

Data-Driven Decision Culture

Transportation authorities should encourage routine use of analytical evidence in operational planning.

Rather than relying solely on historical experience, operational decisions should incorporate measurable performance indicators.

Benefits include:

- Increased transparency.
 - Consistent decision-making.
 - Improved operational accountability.
 - Better allocation of public resources.
-

Continuous Data Collection

Maintaining high-quality operational data is essential for sustaining analytical capability.

Future datasets should continue recording:

- Ticket sales
- Ticket redemption
- Service intervals
- Vessel assignments
- Passenger counts

Additional operational variables such as weather conditions, ferry occupancy, maintenance records, and service disruptions could further enhance future analyses.

6.5 Benefits of the Proposed Analytics System

Implementation of the Ferry Capacity Utilization & Operational Efficiency Analytics System offers several operational advantages.

Operational Benefits

- Improved vessel utilization.
 - Reduced idle operating time.
 - Better congestion management.
 - Enhanced scheduling efficiency.
 - Improved workforce planning.
-

Financial Benefits

- Reduced unnecessary operational expenditure.
 - More efficient deployment of ferry resources.
 - Improved allocation of staffing costs.
 - Better long-term budgeting.
-

Passenger Benefits

- Reduced waiting times.
 - Lower congestion.
 - Improved comfort.
 - More reliable ferry services.
 - Enhanced overall travel experience.
-

Management Benefits

- Real-time operational visibility.
 - Standardized performance measurement.
 - Interactive reporting.
 - Evidence-based planning.
 - Improved strategic decision support.
-

6.6 Future Enhancement Opportunities

The proposed analytical framework can be extended in several ways.

Potential enhancements include:

- Passenger demand forecasting using machine learning.
- Real-time operational monitoring through live ticketing systems.
- Weather-integrated scheduling optimization.
- Predictive congestion alerts.
- Vessel occupancy estimation.
- Fuel consumption optimization.
- Automated operational reporting.
- Geographic visualization of ferry routes.
- Integration with smart transportation infrastructure.

These enhancements would further strengthen operational intelligence and improve long-term ferry system management.

6.7 Overall Discussion

The findings of this research demonstrate that historical ticket transaction data represent a valuable operational asset rather than merely an administrative record.

By combining feature engineering, statistical analysis, operational KPIs, and interactive business intelligence, transportation authorities gain a comprehensive understanding of ferry utilization patterns.

The proposed Ferry Capacity Utilization & Operational Efficiency Analytics System enables decision-makers to transition from reactive operational management toward proactive, evidence-based planning. Such an approach supports efficient resource allocation, improved passenger service, and sustainable public transportation operations.

6.8 Chapter Summary

This chapter translated the analytical findings into practical business insights and strategic recommendations for the Toronto Island Ferry system. The results highlight opportunities to improve scheduling, optimize vessel deployment, manage congestion proactively, reduce idle capacity, and strengthen long-term operational planning.

By adopting a data-driven operational strategy supported by standardized KPIs and interactive dashboards, Toronto Government Parks, Forestry & Recreation can improve service reliability, operational efficiency, and passenger satisfaction while making more effective use of public resources. These recommendations also establish a foundation for future enhancements such as predictive analytics, real-time monitoring, and intelligent transportation planning.

7. Conclusion and Future Work

7.1 Conclusion

This research presented a comprehensive **Ferry Capacity Utilization & Operational Efficiency Analytics System** designed to evaluate and improve the operational performance of the Toronto Island Ferry service using historical ticket sales and redemption data. The study addressed the need for a structured analytical framework capable of transforming routine operational records into actionable insights for transportation planning and decision-making.

A systematic methodology was implemented, beginning with data preprocessing, timestamp standardization, and quality assessment, followed by feature engineering to derive meaningful operational indicators. Key metrics such as the **Capacity Utilization Ratio (CUR)**, **Operational Load Index (OLI)**, **Congestion Pressure Index (CPI)**, **Idle Capacity Indicator**, **Peak Strain Duration**, and **Operational Variability Score** were developed to quantify ferry performance from multiple operational perspectives.

Exploratory Data Analysis (EDA) demonstrated that ferry activity exhibits clear temporal and seasonal patterns. Passenger activity varied significantly across hours, weekdays, weekends, and seasons, confirming that ferry demand is dynamic rather than constant throughout the year. Statistical analyses, including correlation analysis, normality testing, independent sample t-tests, Analysis of Variance (ANOVA), confidence interval estimation, and outlier detection, provided quantitative evidence supporting these operational observations.

The results indicate that congestion is concentrated within recurring operating windows, while extended periods of idle capacity occur during off-peak intervals. These findings

suggest that uniform ferry scheduling throughout the year may not represent the most efficient use of operational resources. Instead, adaptive scheduling strategies informed by historical utilization patterns can improve vessel deployment, optimize staffing, reduce unnecessary operating costs, and enhance passenger service quality.

An important contribution of this research is the development of an integrated set of operational Key Performance Indicators that provide standardized measures of ferry efficiency. Rather than relying solely on ticket transaction totals, transportation managers can evaluate operational performance using multiple complementary indicators that collectively describe utilization, congestion, variability, and operational stability.

To facilitate practical implementation, the analytical outputs were integrated into an interactive **Streamlit dashboard**. The dashboard provides transportation managers with an accessible decision-support platform that includes KPI monitoring, temporal trend visualization, congestion analysis, idle-capacity identification, seasonal comparisons, and downloadable reports. This interface transforms complex analytical outputs into information that can be interpreted and applied quickly within operational environments.

From a management perspective, the proposed analytics system supports evidence-based operational planning by enabling authorities to identify recurring congestion periods, anticipate seasonal demand changes, optimize ferry deployment, and allocate resources more efficiently. The findings demonstrate that historical ticket activity can serve as a valuable operational intelligence resource rather than functioning solely as an administrative record.

Overall, this study successfully demonstrates that data analytics can significantly enhance ferry transportation management by providing measurable, repeatable, and objective assessments of operational performance. The proposed framework supports more informed decision-making while contributing to improved service reliability, passenger comfort, operational efficiency, and sustainable public transportation planning.

7.2 Research Contributions

The study makes several important contributions to transportation analytics and operational management.

Academic Contributions

- Developed a structured analytical framework for evaluating ferry operational efficiency using ticket transaction data.
- Demonstrated the application of feature engineering techniques to derive operational performance indicators.
- Combined exploratory data analysis, statistical testing, and business intelligence within a unified analytical workflow.
- Extended existing transportation analytics research by emphasizing operational efficiency rather than demand forecasting alone.

Practical Contributions

- Introduced standardized KPIs for monitoring ferry performance.
 - Identified congestion-prone and idle-capacity operating periods.
 - Provided operational recommendations for scheduling, staffing, and vessel deployment.
 - Developed an interactive dashboard to support real-time decision-making.
 - Demonstrated how historical ticket records can guide long-term transportation planning.
-

7.3 Limitations of the Study

Although the proposed analytical framework provides valuable operational insights, several limitations should be acknowledged.

First, the analysis relies exclusively on historical ticket sales and redemption data. Other operational variables that may influence ferry performance—such as weather conditions, vessel maintenance schedules, fuel consumption, staffing levels, special events, and service disruptions—were not included because they were unavailable within the dataset.

Second, the Capacity Utilization Ratio represents an analytical estimate derived from ticket activity rather than direct vessel occupancy measurements. Access to vessel-specific passenger capacity and occupancy records would allow more precise utilization calculations.

Third, the study focuses on historical operational analysis rather than predictive modelling. While historical patterns provide valuable planning information, they cannot fully account for unexpected future events such as extreme weather, infrastructure changes, or exceptional tourism fluctuations.

Finally, the findings are specific to the Toronto Island Ferry system. Although the proposed methodology is transferable to other ferry and public transportation systems, implementation may require adjustments to accommodate differences in operational policies, infrastructure, and passenger behavior.

7.4 Future Work

Several opportunities exist to extend the analytical framework developed in this research.

Predictive Analytics

Future studies may incorporate machine learning algorithms to forecast passenger demand, congestion levels, and capacity utilization. Forecasting models could assist transportation managers in proactively adjusting ferry schedules before congestion occurs.

Real-Time Operational Monitoring

Integrating live ticketing data into the Streamlit dashboard would enable continuous operational monitoring. Real-time KPI updates and automated alerts could improve responsiveness during rapidly changing operating conditions.

Weather Integration

Weather conditions significantly influence ferry operations and passenger demand. Future research could combine meteorological data with ticket activity to evaluate weather-related operational impacts and improve scheduling decisions.

Vessel-Level Analytics

If vessel-specific operational data become available, future analyses could compare utilization across individual ferries, enabling more detailed fleet optimization and maintenance planning.

Passenger Flow Modelling

Future research may analyze passenger movement throughout the complete travel journey, including terminal arrival, ticket purchase, boarding, travel time, and disembarkation. Such analyses would support improvements in terminal design and passenger experience.

Geographic Information Systems (GIS)

Integrating Geographic Information Systems (GIS) could provide spatial visualization of ferry routes, terminal operations, and passenger flows. Spatial analytics would enhance infrastructure planning and emergency response strategies.

Artificial Intelligence and Decision Support

Artificial intelligence techniques may be incorporated to automate operational recommendations, optimize scheduling, and support intelligent resource allocation based on continuously updated operational conditions.

7.5 Final Remarks

Efficient public transportation systems depend on accurate operational information and timely decision-making. This research demonstrates that historical ticket transaction data contain valuable operational knowledge that can be transformed into meaningful performance indicators through data analytics.

The Ferry Capacity Utilization & Operational Efficiency Analytics System provides a practical framework for evaluating utilization patterns, detecting congestion, identifying idle capacity, and supporting strategic operational planning. By integrating statistical analysis, feature engineering, visualization, and business intelligence, the proposed system enables transportation authorities to move beyond descriptive reporting toward proactive, evidence-based management.

As public transportation systems continue to adopt digital technologies and data-driven practices, operational analytics will play an increasingly important role in improving efficiency, reducing costs, and enhancing passenger experiences. The methodology developed in this research offers a scalable and adaptable foundation for future transportation analytics initiatives, contributing to more resilient, sustainable, and passenger-focused ferry operations.

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